Digital Sovereignty and the Economic Impact of DNS Centralization

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Abstract. *This paper examines digital sovereignty from an economic perspective, focusing on the centralization of Domain Name System (DNS) services among a few global providers. While existing literature addresses DNS centralization, the economic impacts on national sovereignty have not been explored. This study aims to fill that gap by creating a visualization panel to illustrate the economic distribution of DNS services in Brazilian companies and assessing the financial consequences of DNS failures on Brazil's economy. Key issues include the economic impact of DNS service disruptions and the level of financial dependence of Brazilian firms on foreign providers.*

Resumo. *Este artigo examina a soberania digital a partir de uma perspectiva econômica, com foco na centralização dos serviços de Sistema de Nomes de Domínio (DNS) entre poucos provedores globais. Embora a literatura existente aborde a centralização do DNS, os impactos econômicos sobre a soberania nacional ainda não foram explorados. Este estudo busca preencher essa lacuna criando um painel de visualização para ilustrar a distribuição econômica dos serviços de DNS em empresas brasileiras e avaliando as consequências financeiras das falhas no DNS para a economia do Brasil. As questões-chave incluem o impacto econômico das interrupções nos serviços de DNS e o nível de dependência financeira das empresas brasileiras em relação a provedores estrangeiros.*

1. Introduction

Sovereignty, particularly in the realm of computer networks, has become increasingly important in both academic discussions and the formulation of public policies, as well as in the positioning of individual countries. The study of digital sovereignty, as a contemporary interpretation of an age-old concept, examines various dimensions of computer networks [Azevedo et al. 2024] within a continually evolving international political landscape. Key areas of focus in this field include infrastructure, device manufacturing, intellectual property, standards and regulations, security, and data flow.

In this paper, our objective is to analyze digital sovereignty from an economic perspective, concentrating on the centralization of Domain Name System (DNS) services under the control of a limited number of actors. This phenomenon has been extensively explored in the literature, where previous studies have introduced metrics and mechanisms

to evaluate the market share held by various organizations, including detailed analyses of national contexts in several countries.

However, existing research lacks a specific focus on quantifying a nation's digital sovereignty through the economic and financial implications of DNS centralization. Thus, this work seeks to provide insights into the economic distribution of DNS services among Brazilian companies and to assess the potential financial impact of DNS system failures on the Brazilian economy. For that, the different steps for economic analysis will be defined, and a visualization tool will be developed to depict the economic impacts of DNS centralization.

Consequently, this study is driven by two primary motivations. First, considering the critical role of DNS services in the Brazilian economy, we examine the possible consequences of service failures - whether due to malicious intent or accidental occurrences - on the economy. Second, we propose an analysis of the financial dependence of Brazilian companies on services provided by foreign entities, aiming to measure the cost of these services in relation to the revenue generated by the respective companies. Also, the cause and effects of potential disruptions on such services must be considered.

2. Literature Review

This chapter presents the concepts necessary to understand the work, as well as positioning it in the context of existing works that address digital sovereignty and DNS concentration. First, we list the publications that address the topic and present their contributions to the work. Then, we present the concept of sovereignty in the context of computer networks. We also explain what Domain Name Service is and how its concentration in the hands of a few organizations has been noted.

2.1. Related Work

In [Bates et al. 2021], the authors use three consecutive Distributed Denial-of-Service (DDoS) attacks against the DNS provider Dyn in 2016 as a framework to analyze the resilience of DNS structures, focusing on the increasing concentration of DNS services and the low utilization of redundancy. As part of the study, the author sampled the top 100 U.S. domains under the ".com," ".net," and ".org" TLDs, according to the Alexa Top Sites list. The research spanned the period from 2011 to 2018, with data collected monthly. The study utilized the API provided by [CompleteDNS] to gather historical information on the nameservers associated with each domain. To evaluate market share concentration, the study employed the Herfindahl-Hirschman Index (HHI) [Justice.gov 2024], calculated by squaring the market share of each firm competing in the market and then summing the resulting values.

As a result, the study identified an increase in the HHI over the analyzed period, as seen in Figure 1. One inference made by the author, based on the study, is that the growing outsourcing of services to external providers is one of the factors contributing to the concentration of services among a few companies. During the period of the attacks, Dyn held approximately 10% of the market share.

In another study, [Zembruzki et al. 2022] investigated the global concentration of DNS services using data from the OpenIntel project, which stores information on over 60% of the global domain name space. OpenIntel collects daily data from complete

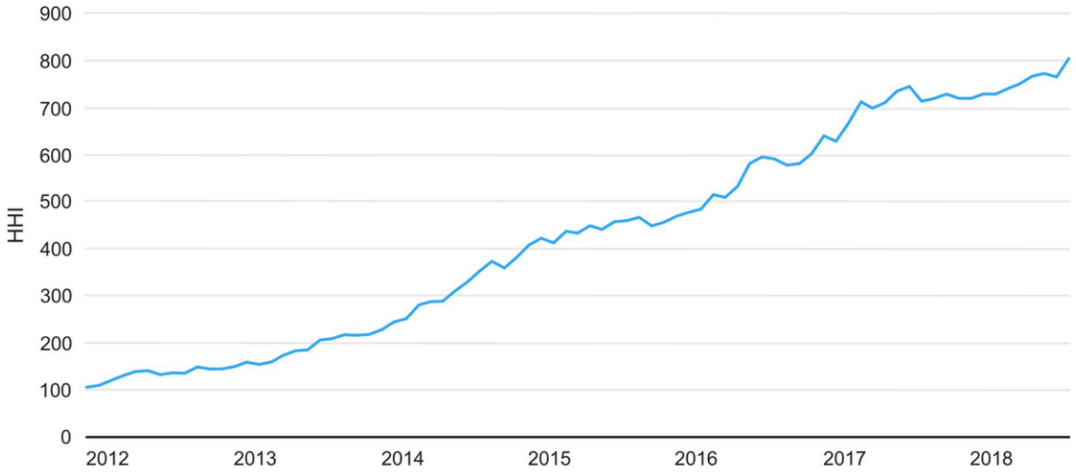


Figure 1. HHI of DNS providers by year [Bates et al. 2021].

zone files of popular gTLDs like ".com", ".net", and ".org", as well as approximately 16 ccTLDs, including Russia (".ru"), Germany (".de"), and China (".cn"). One of the objectives of the research was to analyze a wide range of data, further than in previous works. Therefore, as seen in Table 1 the data set analyzed included 147 to 181 million domains distributed across 11 to 19 TLDs, resolved by approximately 2.4 to 2.8 million IPv4 name servers and approximately 365,000 IPv6 servers, served by approximately 26,000 ASes using IPv4 and 4,500 ASes using IPv6. The study further enriched the analysis by mapping NS IP addresses to ASes using the CAIDA database, linking ASes to organizations and countries for a detailed view of the centralization and geographic dependencies of DNS services.

Table 1. Aggregated datasets, measured yearly [Zembruzki et al. 2022].

Year	TLDs	Domains	NS(v4)	NS(v6)	ASes(v4)	ASes(v6)
2017	11	147.38M	2.84M	364.99k	26.33k	4.11k
2018	13	154.48M	2.68M	331.34k	26.24k	4.37k
2019	14	161.52M	2.41M	214.26k	26.01k	4.54k
2020	17	173.09M	2.44M	244.23k	25.81k	4.74k
2021	19	181.86M	2.42M	242.99k	25.49k	4.93k

The research analyzed the concentration of the DNS market over five years (2017-2021), segmenting domains by year and TLDs, focusing on market share distribution and the geographic location of dominant companies, particularly U.S.-based firms, as seen in Figure 2. In 2017, the five largest DNS ASes resolved 30% of domains using IPv4, increasing to 40% by 2021. Furthermore, the top 100 ASes concentrated approximately 80% of the 1.36 million domains analyzed. These findings reveal a growing reliance on a few DNS service providers, raising concerns about digital resilience and sovereignty due to the risk of systemic failures and over-dependence on limited infrastructure.

To assess national dependence on foreign DNS providers, [Azevedo et al. 2024] analyzed how country-code top-level domains (ccTLDs) are resolved in two major economic blocs: the BRICS nations (Brazil, Russia, India, China, and South Africa) and the European Union.

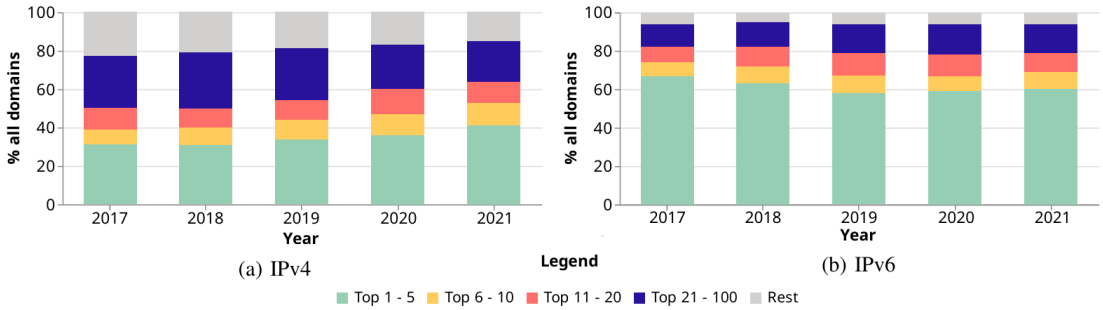


Figure 2. Top DNS Concentration per Year [Zembruzki et al. 2022].

The authors used a measurement approach, previously proposed by [Boeira et al. 2023], to map popular Internet domains to the NS that resolve them, identifying the Autonomous Systems (ASes) responsible and the organizations behind these services. The process is based on DNS record data and Autonomous System (AS) information, such as the most accessed domains list (Tranco List) and the CAIDA database, which provides details about ASes and their associated organizations. Data collection starts with obtaining an updated list of ASes and popular domains. From there, DNS records are retrieved for each domain and mapped to the IP addresses of the name servers. This allows the association of each name server with an AS and, subsequently, with the organization managing it, identifying its geographical location and the regulations it is subject to. The NS resolution flow generated by the approach illustrates the path from the domain to the organization managing the infrastructure. This is crucial for understanding the potential risks of centralization and digital sovereignty, as NS and AS operators can influence DNS resolution operation and security.

Table 2. Information Collected using the measurement approach [Azevedo et al. 2024].

Information	Description	Example
Server Provider	Describe the nameserver allocated to resolve the domain	`ns3.google.com`
Organization Name	Identifies the organization owning the server	Google LLC
AS Number	Identifies the number of the AS managing the infrastructure of server provider	15 169
AS Country	Shows the country where the AS is operating	USA
AS Occurrences	Number of occurrences of the AS number controlling infrastructure of providers per country	30 812 (US) and 32 989 (RU) for BRICS
Centralization Percentage	The percentage that the AS occurrences represents in the data analyzed (*i.e.*, centralization)	35% (US) and 37% (RU) for BRICS

2.2. Digital Sovereignty

The concept of sovereignty in the context of computer networks is typically used to assert some form of collective control over digital content and infrastructures. However, interpretations, subjects, means and definitions of sovereignty can vary significantly, as

contextualized by [Couture and Toupin 2019]. The modern idea of sovereignty, articulated by thinkers like Machiavelli, Bodin, and Hobbes, refers to the supreme authority exercised over a territory or political entity. Until the 20th century, this concept was predominantly applied to control and authority within clearly defined territorial limits. In today's digital age, sovereignty has taken on new importance in public policy formulation and international relations. In the digital realm, the infosphere does not represent a physical territory, data is not finite, and digital assets primarily belong to private companies influenced by market forces and national governments. Therefore, the digital age compels us to reconsider the nature of sovereignty [Floridi 2020].

Various terms related to sovereignty have been analyzed in the digital sphere, including digital sovereignty, cyber sovereignty, internet sovereignty, technological sovereignty, and data sovereignty. [Azevedo et al. 2024] adopts the term "digital sovereignty" as it serves as an umbrella term in the current literature, covering multiple aspects.

As presented by [Floridi 2020], in the realm of computer networks, digital sovereignty refers to the control exercised over the physical layers (resources, infrastructure, devices), code (standards, rules, design), and information (content, data) of digital technologies. This control encompasses the ability to influence and restrict the production, design, use, and outcomes of these technologies.

Countries around the world are making significant efforts to regulate data privacy, enhance cybersecurity, and assert their national sovereignty in the technological sphere. As [Aydn and Bensghir 2019] unlike the traditional conception of sovereignty, digital sovereignty lacks a consolidated international treaty and is approached differently by each nation based on its political and economic priorities.

The urgency of asserting digital sovereignty increased following Edward Snowden's revelations regarding extensive espionage and surveillance conducted by the United States. In the wake of these revelations, political leaders have utilized the concept of digital sovereignty to advocate for reducing technological dependence on foreign nations. A notable case is Brazil: after the Snowden allegations, then-President Dilma Rousseff proposed a plan to diminish the influence of the U.S. and its tech giants on the Brazilian internet. Similarly, Canada has invested in improving its digital infrastructure to reduce the traffic of data passing through US networks [Couture and Toupin 2019]. The growing rivalry between the U.S. and China, along with the COVID-19 pandemic, has further accelerated global efforts to reduce technological and economic dependencies.

The European Union has been actively working to safeguard its digital sovereignty and to achieve strategic technological autonomy [Barrinha and Christou 2022]. It addresses digital sovereignty through both technical and regulatory measures. On the technical side, this includes ensuring data confidentiality and integrity via standards-compliant hardware, the adoption of open-source software, and localized data processing and storage through the European Federated Cloud Initiative. On the regulatory side, it employs the General Data Protection Regulation (GDPR) [Tipper and Krishnamurthy 2022].

2.3. Domain Name System

Virtually all modern Internet applications depend on the Domain Name System (DNS) to function properly. Its main function, translating names into IP addresses, is essen-

tial, especially for users, who are accustomed to recognizing “locations” on the internet through friendly words, titles, and abbreviations, rather than numeric IP addresses [Deccio et al. 2009].

Although the concept of name resolution is relatively straightforward, the system that implements it is highly complex and has broad impacts. Often, the name resolution of a domain depends on servers that are outside the control of the domain owner and are managed by third parties [Deccio et al. 2009]. The large scale of the database and the frequency of its updates make it essential to maintain this system in a distributed manner [Mockapetris 1987].

According to [Tanenbaum et al. 2021], the management of the top of the Internet’s naming hierarchy is carried out by the Internet Corporation for Assigned Names and Numbers (ICANN). The conceptual structure of the Internet is organized into top-level domains (Top-Level Domains - TLDs), each housing several hosts. These TLDs are subdivided into subdomains, which can be continuously divided, forming a tree-like naming hierarchy, whose leaves represent domains that do not have subdomains, as seen in Figure 3.

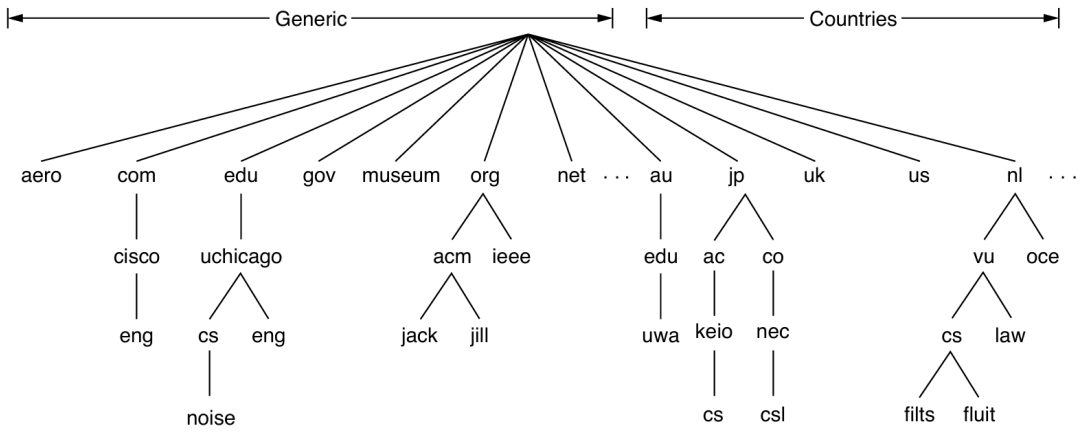


Figure 3. Portion of DNS name space [Tanenbaum et al. 2021].

A Top-Level Domain (TLD) is a zone located immediately below the root of the DNS hierarchy, exemplified by extensions such as ”.com” or ”.jp”. TLDs are often classified into subgroups, such as Country Code Top-Level Domains (ccTLDs), which represent specific countries or regions (e.g., ”.br” for Brazil), and Generic Top-Level Domains (gTLDs), which encompass generic categories such as ”.org” or ”.edu” [Hoffman et al. 2019].

TLDs are operated by companies called registries, designated by ICANN. For example, Verisign is responsible for registering the ”.com” domain. At a lower hierarchical level, registrars are responsible for selling domain names directly to end users. The market has several registrars, which compete with each other by offering different prices and services. Popular examples are: Domain.com, GoDaddy and NameCheap [Tanenbaum et al. 2021]. In Brazil, Registro.br is responsible for the registration and maintenance of domain names that use the .br [Registro.br].

The DNS namespace is organized hierarchically. Zones represent parts of this

namespace, are managed by a single entity, and are delegated to organizations in a descending manner, starting with the root zone. In the resolution process, name servers (resolvers) send queries to authoritative servers, which can provide a direct answer or a referral to a delegated zone [Deccio et al. 2009]. Based on data obtained from OpenIntel, between 2017 and 2021, [Zembruzki et al. 2022] quantified the numbers of TDLs, domains, NSs and ASs, for both IPv4 and IPv6.

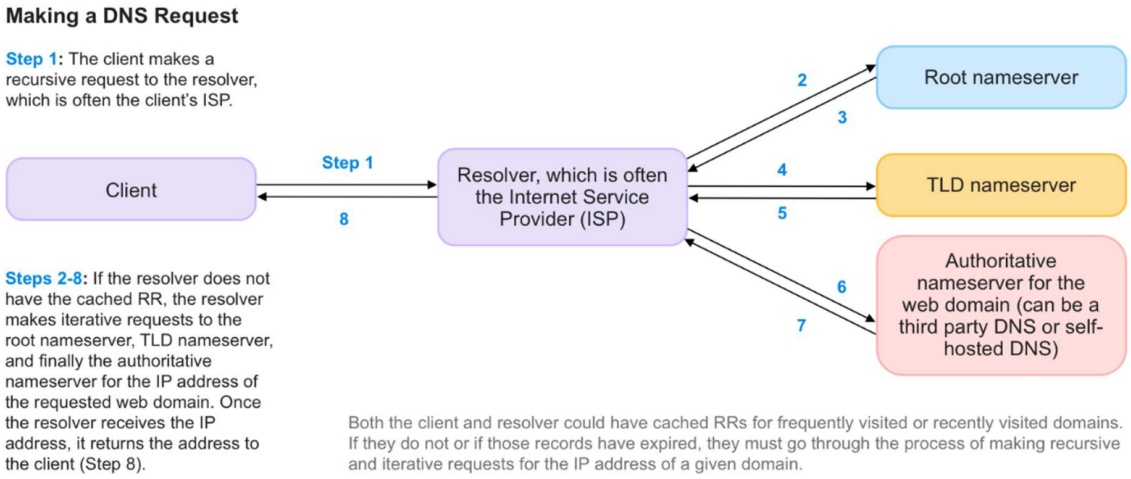


Figure 4. Process of a DNS request [Bates et al. 2021].

As presented by [Azevedo et al. 2024], an Autonomous System (AS) is a network of interconnected computing devices that operate under the same policy. It is typically managed by a single entity, such as an Internet Service Provider (ISP) or a technology organization, and is identified by an Autonomous System Number (ASN). Therefore, it is possible to associate the IP address of any Name Server (NS) with an AS, and consequently, with its operator and geographical region.

2.4. DNS Centralization

According to [Moura et al. 2020], in recent years, there has been a growing concern about the centralization of the Internet, which consists of the concentration of services, infrastructure and users in a few companies. [Wang et al. 2021] when analyzing the 10 thousand most accessed sites on the Tranco list, concludes that the companies Cloudflare and Amazon alone concentrate more than 30% of DNS domains, as seen in Figure 5.

When analyzing data from the [OpenINTEL] project, and adding it to the prefix-to-AS data and AS-to-organization from CAIDA [CAIDA 2013] [CAIDA 2014], [Zembruzki et al. 2022] obtained an overview of the evolution of DNS centralization over the period from 2017 to 2021. The study concludes that in 2021 the concentration of domains using IPv4 from the top 5 DNS ASs reached the 40% mark, with 80% of the domains concentrated in the top 100 ASes of DNS providers.

Such concentration generates vulnerability, as presented by [Moura et al. 2020], as it creates points of dependence susceptible to attacks, such as the Denial-of-Service (DoS) attacks against DNS service providers that occurred in 2016, harming Dyn [Hilton 2016], [Chacos 2016], and in 2019, harming AWS [Williams 2019]. In both cases, important websites and services experienced outages.

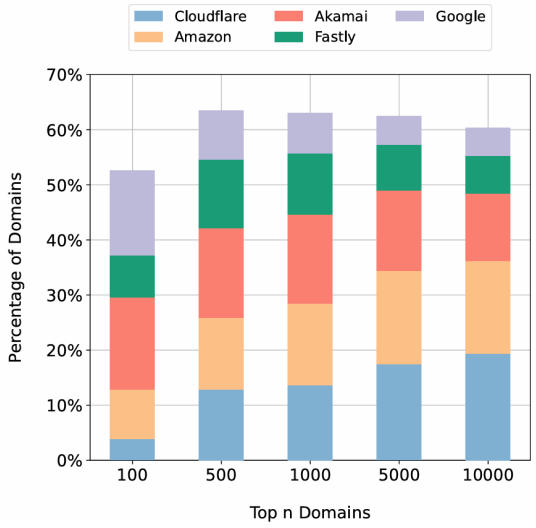


Figure 5. Percentage of domains that use exclusively one AS organization to host their index page [Wang et al. 2021].

A DNS outage occurs when a domain name cannot be resolved to its corresponding IP address. During this time, clients will send a DNS query for a domain name, but the DNS resolver may respond with an outdated IP address from its cache, which will not be functional. Alternatively, if the resolver attempts to query the authoritative name server for the domain and does not receive a response, the resolution will fail [Pramatarov 2024].

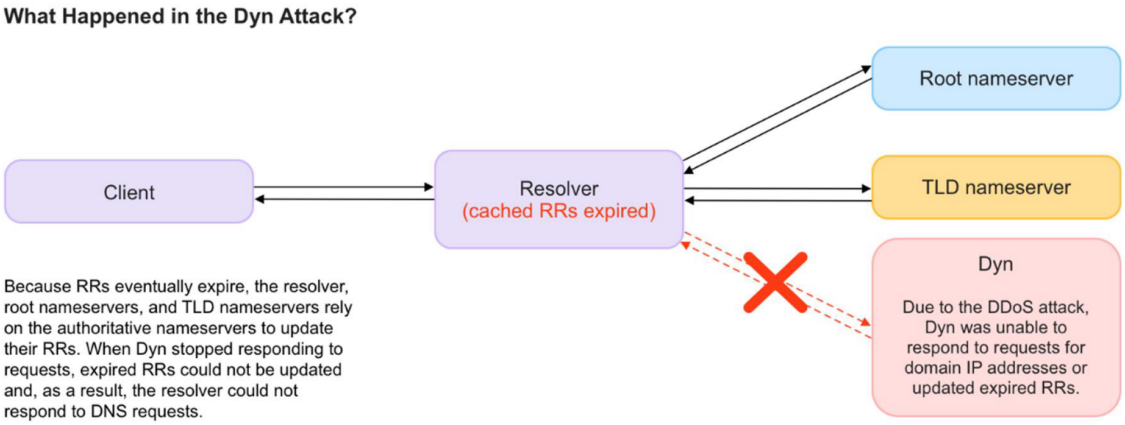


Figure 6. WHat happened in the Dyb attack [Bates et al. 2021].

The main causes of DNS failures include Distributed Denial of Service (DDoS) attacks, maintenance of authoritative name servers, issues in the data center hosting the authoritative name server, delays in DNS propagation, or misconfigurations.

3. Proposed Work

This section presents the methodology used to develop the work, detailing the steps taken to achieve the objectives. The methodology is divided into two main parts: the definition of the approach for data collection and the development of the visualization panel and financial impact analysis.

3.1. Data Collection

The data will be collected using a pipeline similar to the one proposed by [Azevedo et al. 2024], which determines the entire flow from the domain name to the organization handling the AS that manages the IP of the associated NS. The process begins with obtaining lists of popular ASes and domains, from which the DNS records for each domain are retrieved.

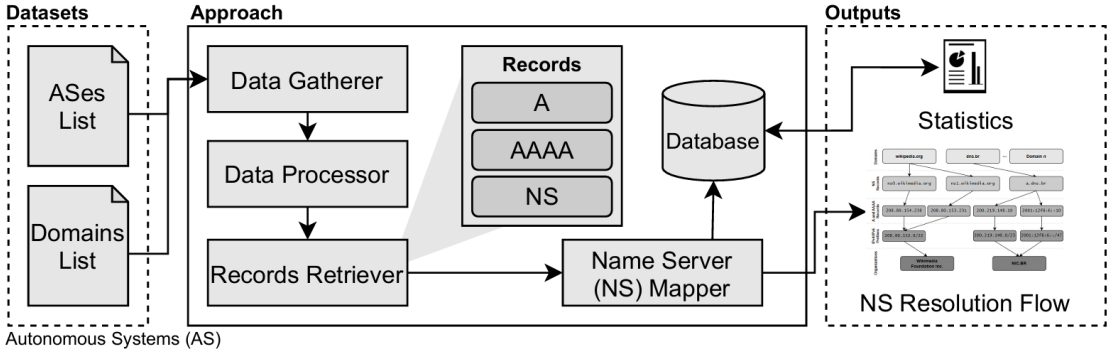


Figure 7. Overview of the Approach to Analysis Domains [Azevedo et al. 2024].

The DNS records are then mapped to the IPs of the name servers, associating them with specific ASes. This makes it possible to identify the organization that manages each AS. As a result, the pipeline generates an NS resolution flow, which traces the resolution path of a domain to the organization responsible for the DNS infrastructure, allowing the analysis of risks related to the centralization of servers and the security of DNS resolution. The process is implemented in Python, using the dnspython library to manipulate DNS records. The collected data is stored in a database, allowing the processing and analysis of data necessary for the subsequent stages of this work.

The data will be collected from the OpenIntel project, which provides information on the DNS hierarchy, and from the CAIDA project, which provides information on the AS. Additionally, financial information will be added to the pipeline, such as the revenue generated by Brazilian companies that rely on DNS services provided by foreign entities, as well as the cost of these services.

3.2. Financial Impact Analysis

The financial impact analysis will be conducted using the revenue generated by Brazilian companies that rely on DNS services provided by foreign entities, as well as the cost of these services. The model will simulate the financial consequences of DNS failures on the Brazilian economy, taking into account the number of companies affected, the duration of the outage, and the revenue lost due to the outage.

4. Schedule

The work will be carried out over the next six months, or 24 weeks, and is divided into the following tasks:

- Literature review and definition of the range of scope, weeks 1 to 3.
- Investigation of causes of DNS failures, weeks 3 to 5.

- Definition of key metrics for economic and financial analysis of DNS, weeks 5 to 7.
- Development of the pipeline for data collection, weeks 7 to 16.
- Development of the visualization panel and analysis of the results, weeks 10 to 18.
- Writing of the thesis, weeks 10 to 20.
- Final adjustments and submission of the work, weeks 21 to 24. .

5. Conclusion

This work aims to analyze digital sovereignty from an economic perspective, focusing on the centralization of DNS services among a few global providers. While existing literature addresses DNS centralization, the economic impacts on national sovereignty have not been explored. This study aims to fill this gap by creating a visualization panel to illustrate the economic distribution of DNS services in Brazilian companies and to assess the financial consequences of DNS failures on the Brazilian economy. Key issues include the economic impact of DNS service disruptions and the level of financial dependence of Brazilian firms on foreign providers.

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